

Byeong-Il Ham

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Education

Korea Advanced Institute of Science and Technology , Robotics	Daejeon, KR 2024 – present
Korea Advanced Institute of Science and Technology , Robotics Robust Swing Leg Planning Algorithm for Quadruped Robots Using Dual Convex Model-Predictive Control	Daejeon, KR 2022 – 2024
University of Kwangwoon , School of Robotics	Seoul, KR 2016 – 2022

Experience

IEEE , Graduate Student Member	Mar 2025 – present 1 year 3 months
IEEE Robotics and Automation Society , Technical Committee Member - Model-Based Optimization for Robotics - Humanoid Robotics - Robot Control - Robot Learning	Jan 2026 – present 5 months

Awards

Best System Paper Award Hyun-Bin Kim, Chanseok Kim, Byeong-Il Ham, Jeonguk Kang, Minseong Choi, Keun-Ha Choi, Kyung-Soo Kim	Dec 2024
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Publications

Dual-MPC Footstep Planning for Robust Quadruped Locomotion Byeong-Il Ham, Hyun-Bin Kim, Jeonguk Kang, Keun Ha Choi, Kyung-Soo Kim	
Ground-Aware Octree-A* Hybrid Path Planning for Memory-Efficient 3D Navigation of Ground Vehicles Byeong-Il Ham, Hyun-Bin Kim, Kyung-Soo Kim ieeexplore.ieee.org/abstract/document/11301310	
Adaptive Gait Pattern Switching under External Disturbances Using Multi-Modal MPC Jeonguk Kang, Byeong-Il Ham, Seungho Han, Hyun-Bin Kim, Kyung-Soo Kim ieeexplore.ieee.org/document/11217651	
Compact and Optimized Optical Six-Axis Force/Torque Sensor for Legged Robots Hyun-Bin Kim, Byeong-Il Ham, Keun-Ha Choi, Kyung-Soo Kim ieeexplore.ieee.org/document/11194133	
A Miniature High-Resolution Tension Sensor Based on a Photo-Reflector for Robotic Hands and Grippers Hyun-Bin Kim, Byeong-Il Ham, Kyung-Soo Kim ieeexplore.ieee.org/abstract/document/11150551	

Development of Remote Piping Inspection System with Dual-Mode Locomotion Quadruped Robot

Hyun-Bin Kim, Chanseok Kim, Byeong-Il Ham, Jeonguk Kang, Minseong Choi, Keun-Ha Choi, Kyung-Soo Kim

link.springer.com/chapter/10.1007/978-3-031-92011-0_25

Temperature Compensation Method for a Six-Axis Force/Torque Sensor Using a Gated Recurrent Unit

Hyun-Bin Kim, Seokju Lee, Byeong-Il Ham, Keun-Ha Choi, Kyung-Soo Kim

ieeexplore.ieee.org/document/11021311

External force adaptive control in legged robots through footstep optimization and disturbance feedback

Jeonguk Kang, Hyun-Bin Kim, Byeong-Il Ham, Kyung-Soo Kim

ieeexplore.ieee.org/document/10731682

Projects

Development of an AI Robot Platform and Standard System for Seawater Pipeline Inspection

Apr 2022 – Dec 2022

Developed robot platform technologies and standard systems for seawater pipeline inspection.

- Korea Hydro Nuclear Power
- Embedded Systems Engineering

Development of high-speed precision driving algorithm for intelligent active protection technology

June 2023 – Dec 2023

Development of active protection technology system.

- Hanwha Aerospace
- Project Manager

DRB-KAIST SketchTheFuture Research Center

July 2025 – present

Robotic immersive interaction platform for future mobility space design.

- DRB-Industrial
- Project Manager since 2025

Single-Person Multi-Purpose Autonomous Micro-Mobility for Factory and Campus

Feb 2024 – present

Developed autonomous micro-mobility platforms and embedded control systems for factory and campus environments.

- DRB-Industrial
- Project Manager

Skills

Tools

Interests

Robotics

Languages

Korean

Native speaker

English

Intermediate